State Estimation for Connected and Autonomous Vehicles under Unknown V2V Communication Delays[⋆,⋆⋆]

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ABSTRACT

This paper addresses the problem of state estimation for Connected and Autonomous Vehicles (CAVs) operating under Cooperative Adaptive Cruise Control (CACC), where the longitudinal state of a leading vehicle must be reconstructed from Vehicle-to-Vehicle (V2V) communication subject to time-varying packet-level delays. A defining feature of V2V channels is that the delay $\tau(t)$ is intrinsically *unknown*: clock synchronization protocols (NTP, PTP, GPS-disciplined) provide an estimate $\hat{\tau}(t)$, but with a non-zero residual error and possibly under adversarial timestamp manipulation. Existing observer-based methods for delayed measurements assume that $\tau(t)$ is exactly known and break down under this realistic uncertainty. We propose a state observer design for CAVs that explicitly accounts for the unknown V2V delay, by extending the recent dynamic-extension framework of Arezki, Zemouche and Bagnerini (2026) to the unknown-delay setting. As a secondary theoretical contribution, we derive a general-purpose observer for nonlinear systems with delayed nonlinear outputs and unknown delay, certifying input-to-state stability (ISS) of the estimation error with respect to the delay-estimation error $\delta\tau(t) \triangleq \tau(t) - \hat{\tau}(t)$. The synthesis is reduced to a bicriterion semidefinite program (SDP) that jointly optimizes the maximum tolerable delay and the ISS gain. The framework is validated on a CACC scenario in highway driving conditions, and the ISS bound is exploited as a residual-based detector for V2V timing-channel cyber-attacks.

1. Introduction

Connected and Autonomous Vehicles (CAVs) rely on Vehicle-to-Vehicle (V2V) communication to extend their perception beyond on-board sensors and to coordinate longitudinal motion in real time. A canonical instance is Cooperative Adaptive Cruise Control (CACC), in which a follower vehicle receives the longitudinal state of its predecessor (position, velocity, and possibly acceleration) through V2V messages, and uses this information to maintain a tight inter-vehicular gap with provable string stability (Ploeg, Shukla, van de Wouw and Nijmeijer, 2014). CACC and its multi-vehicle generalization, the platoon, are central building blocks of cooperative driving and are expected to deliver substantial gains in road throughput, energy efficiency, and safety.

A fundamental challenge in CACC, however, is that V2V messages are subject to non-deterministic communication delays $\tau(t)$ caused by channel access (DSRC contention, C-V2X scheduling), congestion, retransmissions, and processing latency. These delays can range from tens to several hundreds of milliseconds, and they are typically *unknown* at the receiver: only an estimate $\hat{\tau}(t)$ can be obtained, either through timestamping (limited by clock synchronization residuals of the order of a millisecond with NTP, sub-millisecond with GPS-disciplined oscillators) or through cross-correlation of acceleration signals (when the follower has on-board radar/lidar sensing of the leader). Furthermore, recent work on connected-vehicle cyber-security (Jeon, Xie, Zemouche and Rajamani, 2020) has shown that V2V timestamps are an attack surface: an adversary can inject falsified $\hat{\tau}(t)$ values to corrupt the receiver's state estimate without altering the message payload.

The state-of-the-art literature on observer design for delayed measurements (Germani, Manes and Pepe, 2002; Van Assche, Ahmed-Ali, Hann and Lamnabhi-Lagarrigue, 2011; Cacace, Germani and Manes, 2014; Khosravian, Trumpf and Mahony, 2015; Zemouche, Zhang, Mazenc and Rajamani, 2019; Adil, Hamaz, Ndoye, Zemouche, Laleg-Kirati and Bedouhene, 2022; Arezki et al., 2026) assumes that $\tau(t)$ is exactly known in real time. This assumption is incompatible with realistic V2V scenarios. Treating $\tau(t)$ as known when it is in fact estimated exposes the observer to a model mismatch whose effect on the estimation accuracy must be characterized.

Contributions. This paper addresses precisely this gap, with both an application-driven and a methodological contribution:

- **(C1)** We formulate the state estimation problem for a CAV operating under CACC with V2V delays, identify the unknown-delay nature of the problem, and characterize the available delay-estimation mechanisms (timestamping, cross-correlation). The leading-vehicle longitudinal model is cast in companion form, and the V2V output is treated in both linear (BSM-based) and nonlinear (vision-based time-to-collision) variants.
- **(C2)** As a methodological tool to solve this problem, we extend the dynamic-extension observer framework of Arezki et al. (2026) to the unknown-delay case. The integral term of the observer is evaluated over the *estimated* delay window $[t - \hat{\tau}(t), t]$, and the LMI synthesis condition is enriched by a structural margin

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in the direction of the perturbation, derived through an exact completion of squares. The maximum tolerable delay $\tau^\star$ is preserved with respect to the known-delay case: robustness to $\delta\tau$ is paid for solely through the new structural block of the LMI.

(C3) The observer is synthesized via a bicriterion SDP that jointly optimizes the delay tolerance and the ISS gain. The framework is validated on a representative CACC scenario, and the ISS bound is shown to provide a residual-based detector for V2V timing-channel attacks, naturally extending the cyber-monitoring framework of Jeon et al. (2020).

Paper organization. Section 2 formulates the CAV state estimation problem under unknown V2V delay, presents the longitudinal model of the leading vehicle, the V2V output, and the available delay estimators, and casts the design objective as an explicit problem statement. Section 3 develops the general-purpose theoretical solution, namely the unknown-delay extension of the dynamic-extension observer framework, and derives the LMI-based synthesis condition together with the ISS bound. Section 4 returns to the CACC application, instantiates the design on the leading-vehicle model, and reports simulations under nominal and adversarial V2V conditions. Section 5 concludes.

2. State Estimation in CACC under Unknown V2V Delay

This section describes the CAV scenario considered in this paper: a follower vehicle equipped with V2V communication and a state observer must reconstruct the longitudinal state of a leading vehicle whose position is broadcast over a delayed channel. The delay is unknown, but an external estimate is available. The section establishes the system model, the output model, the delay-estimation model, and finally states the problem to be solved.

2.1. Longitudinal model of the leading vehicle

We consider a two-vehicle CACC scenario, as shown in Fig. 1: a leading vehicle (*leader*) drives ahead in the same lane as a following vehicle (*follower*), which carries the state observer designed in this paper. The leader broadcasts its longitudinal state through a wireless V2V communication channel, and the follower seeks to reconstruct the leader's state in real time despite the unknown communication delay. In this paper we focus on designing the observer to estimate the leader's state under the unknown delay based on the longitudinal model; the lateral dynamics (steering, lane change) are out of scope.

Let the leader's longitudinal state vector be

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} \in \mathbb{R}^3, \tag{1}$$

where $x_1(t)$ [m] is the longitudinal position of the leader's center of mass in a fixed inertial road frame, $x_2(t)$ [m/s] is

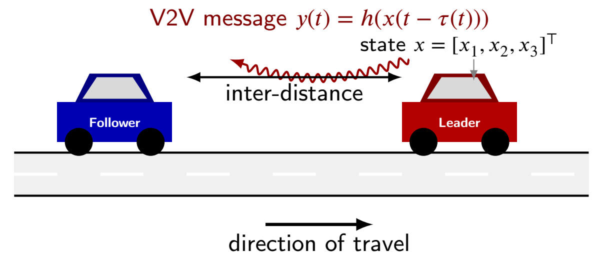

Figure 1: The CACC two-vehicle scenario considered in this paper. The leader's longitudinal state $x(t) = [x_1, x_2, x_3]^\top$ (position, velocity, acceleration) is transmitted over a delayed V2V wireless channel; the follower estimates $x(t)$ from the delayed measurement $y(t)$ and the available delay estimate $\hat{\tau}(t)$.

its longitudinal velocity, and $x_3(t)$ [m/s^2] is its longitudinal acceleration.

Modern vehicles do not realize a commanded acceleration instantaneously. The powertrain (engine, throttle, transmission, brake actuators) introduces a finite response time. The standard first-order approximation, widely used in the vehicle with CACC (Ploeg et al., 2014), models the actual acceleration $x_3(t)$ as

$$\dot{x}_3(t) = -\tfrac{1}{\eta}\, x_3(t) + \tfrac{1}{\eta}\, u(t) - \tfrac{1}{m} F_{\mathrm{res}}(x(t)), \tag{2}$$

where $u(t)$ [m/s^2] is the desired (commanded) acceleration generated by the leader's own controller, $\eta > 0$ [s] is the powertrain time constant, and $F_{\mathrm{res}}(x)$ [N] aggregates the road-load resistive forces.

The dominant resistive force on a flat road is aerodynamic drag, which can be modeled as:

$$F_{\mathrm{res}}(x) \;\approx\; \tfrac{1}{2}\rho_{\mathrm{air}} C_d A_f\, x_2^2(t) \;=\; \tilde{c}_d\, x_2^2(t), \tag{3}$$

where ρ_{air} [kg/m^3] is the air density, C_d [-] the drag coefficient (typically 0.25–0.35), and A_f [m^2] the frontal area. The lumped coefficient is $\tilde{c}_d \triangleq \tfrac{1}{2}\rho_{\mathrm{air}} C_d A_f$. Rolling resistance and grade-induced gravity components, both linear or constant in x, are neglected at the modeling stage and could be absorbed into $u(t)$ if needed without changing the structure of (4) below. Substituting (3) into (2), the full longitudinal model is

$$\dot{x}(t) = \begin{bmatrix} x_2(t) \\ x_3(t) \\ -\tfrac{1}{\eta}\, x_3(t) - \tfrac{c_d}{\eta\, m}\, x_2^2(t) + \tfrac{1}{\eta}\, u(t) \end{bmatrix}, \tag{4}$$

where we set $c_d \triangleq \eta\, \tilde{c}_d$ to absorb the powertrain time constant in the drag coefficient.

The longitudinal model (4) has typically *(triangular) form*,which means the first two state equations are pure integrators, and all the nonlinearity (the quadratic drag $-c_d x_2^2/(\eta m)$) and the control input $u(t)$ appear together in the last coordinate $\dot{x}_3$. This is precisely the structure exploited by the high-gain observer Arezki et al. (2026). Consequently, we can rewrite (4) in the triangular form:

$$\dot{x}(t) = A\, x(t) + B \left[f_x(x(t)) + \tfrac{1}{\eta} u(t) \right], \tag{5}$$

with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad f_x(x) = -\frac{1}{\eta} x_3 - \frac{c_d}{\eta m} x_2^2,$$

which fits exactly the framework of (Arezki et al., 2026, Eq. (30) and Remark 11).

In the real world, the velocity is limited by 40 m/s ($\approx$ 144 km/h) and the the commanded acceleration is limited by 4 m/s^2 covering all comfort-bounded acceleration/deceleration profiles. Considering the physically bounded velocity range and actuator authority in the vehicle, we consider the state and input constraints as

$$|x_2(t)| \leq \bar{v}, \qquad |u(t)| \leq \bar{u}, \qquad \forall t \geq 0, \tag{6}$$

2.1.1. *Interpretation of the envelope assumption.*

Three remarks are in order on (6):

(i) Restricting x_2 to $|x_2| \leq \bar{v}$ is *not* a control-theoretic limitation: it merely encodes the physical fact that highway-driven vehicles do not exceed a known speed limit. Mathematically, it allows applying a Hilbert-projection-based Lipschitz extension (Zemouche and Rajamani, 2022) to f_x, recovering global Lipschitzness with constant $\gamma_{f_x} = 2\bar{v}\, c_d/(\eta m)$.

(ii) Restricting u to $|u| \leq \bar{u}$ corresponds to the comfort-driven authority bound used by the leader's own controller. Together with (i), this provides the uniform bound on $g(\hat{\zeta}, u)$ required by the unknown-delay observer of Section 3 (see Assumption 1 and the explicit computation of M_g in Section 4.2).

(iii) Both bounds are easily verified at run-time (vehicle CAN bus, V2V message envelope), so the values $\bar{v}$ and $\bar{u}$ are known design parameters. The observer is robust outside the envelope only in the local-Lipschitz sense; entering a regime $|x_2| > \bar{v}$ in flight would invalidate the synthesis a posteriori, but this regime is not encountered in normal CACC operation.

2.2. V2V delayed output measurement

The follower vehicle reconstructs the leader's state from a single delayed measurement $y(t)$. Whatever the physical sensing chain that produces $y(t)$, the structure of the measurement is always

$$y(t) \;=\; h\big(x(t - \tau(t))\big), \tag{7}$$

where $h(\cdot)$ is the output function (linear or nonlinear, depending on the sensing modality) and $\tau(t) \geq 0$ is the unknown V2V end-to-end delay. The composition is what makes the problem nontrivial: even if h were the identity, the time argument $t - \tau(t)$ would still be unknown to the receiver, since $\tau(t)$ is itself unknown. Throughout this paper we consider two complementary instances of (7), denoted

(O1) *linear V2V output*

(O2) *nonlinear vision-augmented output*

They cover the two practical sensing configurations encountered in modern CAV deployments.

2.2.1. *Anatomy of the V2V delay.*

Before describing the two output cases, we briefly anatomize the V2V delay $\tau(t)$ itself, since both cases share the same delay model. Fig. 2 represents the five physical contributions that aggregate into the end-to-end delay:

(i) sensor sampling and on-board state encoding at the leader (typically deterministic, $\sim$ 1 ms);

(ii) medium-access contention or scheduling at the leader's V2V transmitter (DSRC: random back-off, $\sim$ 1–10 ms; C-V2X: scheduled, $\sim$ 1–5 ms);

(iii) radio propagation (negligible at vehicular distances, $< 10\,\mu$s);

(iv) reception and decoding at the follower's V2V receiver, including possible retransmissions in case of CRC failure ($\sim$ 1–50 ms);

(v) buffering and delivery to the on-board observer's clock domain ($\sim$ 1–10 ms).

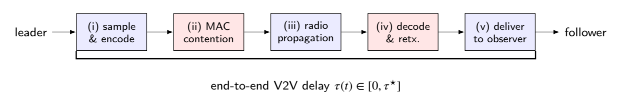

Figure 2: Anatomy of the end-to-end V2V delay $\tau(t)$. Stages (ii) and (iv) (in red) are the dominant random contributions and are responsible for the time-varying nature of $\tau(t)$. Stages (i), (iii), (v) (in blue) are essentially deterministic. The total $\tau(t)$ is bounded by a known upper bound $\tau^\star$ that depends on the V2V technology (DSRC vs C-V2X) and on the channel-congestion regime.

The aggregated delay is therefore a positive, bounded, time-varying signal of the form

$$0 \leq \tau(t) \leq \tau^\star, \qquad \tau^\star \in [50 \text{ ms}, 500 \text{ ms}], \tag{8}$$

the upper bound depending on the V2V technology and the level of channel congestion (Jeon et al., 2020). Three properties of $\tau(t)$ should be emphasized for the observer design that follows:

(P1) $\tau(t)$ **is unknown at the follower side.** Stages (ii) and (iv) involve random back-offs and possible retransmissions over a shared channel; their realization is not deducible from any data available at the receiver.

(P2) $\tau(t)$ **is bounded by a known** $\tau^\star$**.** Standard V2V profiles (SAE J2735, ETSI ITS-G5) guarantee an upper-bound latency negotiated at the protocol level; this $\tau^\star$ is a design-time parameter.

(P3) $\tau(t)$ **varies on a time scale of tens of milliseconds.** It is therefore reasonable to model $\tau(t)$ as piecewise constant or slowly varying compared to the vehicle dynamics (whose dominant time scale is the powertrain constant $\eta \sim 0.5$ s).

Properties (P1)–(P2) are the key reason why the observer of Section 3 requires only an estimate $\hat{\tau}(t)$ of the delay (Section 2.3) and only the upper bound $\tau^\star$, not the exact value.

2.2.2. *Output case (O1): Linear V2V output via BSM.*

The simplest and most widespread instance corresponds to the leader broadcasting its position x_1 through Basic Safety Messages (BSM, defined in SAE J2735). At a nominal broadcast rate of 10 Hz, each BSM carries a packet $\langle t_s, x_1(t_s), \dots \rangle$ where t_s is the leader's transmission timestamp; the follower receives this packet at the local time $t = t_s + \tau(t)$, hence the linear delayed output

$$y(t) \;=\; x_1\big(t - \tau(t)\big) \;=\; C_x\, x\big(t - \tau(t)\big), \qquad C_x \;=\; \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \tag{9}$$

The output function is linear in the state and depends only on x_1. Fig. 3 schematizes this measurement chain.

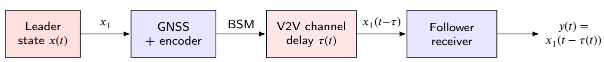

Figure 3: Measurement chain for the linear V2V output (O1). The leader's GNSS receiver samples the position x_1 and packs it into a BSM, which is transmitted over the V2V channel and incurs the unknown delay $\tau(t)$. The follower decodes the message and delivers $y(t) = x_1(t - \tau(t))$ to the state observer.

Interpretation. Case (O1) corresponds to the standard CACC sensing assumption used by the majority of the V2V control literature. The output map $h(x) = C_x x$ is a globally Lipschitz linear functional with constant $\gamma_h = 1$, and its derivative $\partial h/\partial x = C_x$ does not depend on x, which simplifies the verification of the Lipschitz/boundedness assumptions in Section 4.2. In particular, the function $g(\zeta, u) = (\partial h/\partial x)(x)\, f(x, u) = x_2$ in this case is unbounded only if the velocity is unbounded, hence $M_g = \bar{v}$ from (6).

2.2.3. *Output case (O2): Nonlinear vision-augmented output.*

Modern CAVs increasingly fuse V2V data with on-board perception sensors, in particular forward-looking monocular cameras. Such cameras do not measure the leader's position directly: they extract *visual cues* (image-plane bounding-box size, expansion rate, optical flow) which decode into pseudo-distances. A widely used cue is the *inverse time-to-collision* (ITTC), proportional to $1/(\Delta x_1 + d_0)$ where Δx_1 is the inter-distance and $d_0 > 0$ is a safety offset that prevents the singularity at zero distance and absorbs the geometric calibration of the camera. When the leader's V2V message is augmented by such an on-board perception cue evaluated at the same delayed instant, the follower's effective output becomes

$$y(t) \;=\; h\big(x(t - \tau(t))\big), \qquad h(x) \;=\; \frac{1}{x_1 + d_0}. \tag{10}$$

The output function is nonlinear (rational in x_1), monotone decreasing, smooth, and Lipschitz on $\{x_1 \geq 0\}$ with constant $\gamma_h = 1/d_0^2$. Its derivative,

$$\frac{\partial h}{\partial x}(x) \;=\; \begin{bmatrix} -\dfrac{1}{(x_1 + d_0)^2} & 0 & 0 \end{bmatrix}, \tag{11}$$

is itself bounded and Lipschitz on the operating envelope $\{x_1 \geq 0\}$, ensuring that the function $g(\zeta, u) = (\partial h/\partial x)(x)\, f(x, u)$ entering the augmented observer of Section 3 is also Lipschitz. Fig. 4 schematizes this fusion chain.

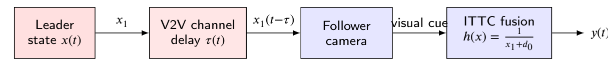

Figure 4: Measurement chain for the nonlinear vision-augmented output (O2). The follower's camera observes the leader at the delayed instant (the perceptual delay of camera processing is small and is absorbed into the V2V delay $\tau(t)$ for analysis purposes), and the ITTC fusion module produces the effective output $y(t) = 1/(x_1(t - \tau(t)) + d_0)$.

Interpretation. Case (O2) is more demanding than (O1) for two reasons. First, the output map h is nonlinear, so the receiver cannot simply linearize and decode x_1 from y; it must invert a nonlinear relation, which is precisely the difficulty addressed by the dynamic-extension framework of Section 3. Second, the constant $\gamma_h = 1/d_0^2$ becomes large for small d_0, which translates into a stronger sensitivity of the ISS gain κ in Theorem 1; the safety offset d_0 is therefore both a physical regularizer (avoiding $h \to \infty$ as $x_1 \to 0$) and a design knob trading off perceptual sensitivity against robustness.

2.2.4. *Why both cases matter.*

The two cases (O1) and (O2) are not redundant: they correspond to two distinct deployment regimes of CAVs. Case (O1) covers the V2V-only sensing typical of early-deployment CACC, where the follower has no forward camera and relies entirely on broadcasted positions. Case (O2) covers the fused V2V+vision sensing of modern Level-3/4 autonomous platforms, where on-board perception is the primary sensing modality and V2V acts as a redundant or corroborative source. The unified framework of Section 3 handles both cases through a single observer structure, the only difference between (O1) and (O2) being the analytic form of h and of the resulting Lipschitz constants in the LMI synthesis.

2.3. Estimation of the unknown delay $\tau(t)$

The previous subsection has established that the unknown delay $\tau(t)$ in either (9) or (10) is bounded but not directly measurable. In a realistic CAV deployment, however, the follower platform always has *some* information about $\tau(t)$, obtained through a dedicated subsystem that is structurally separate from the state observer. This subsystem produces an estimate $\hat{\tau}(t)$, which we will use as an input to the state observer designed in Section 3. We denote the residual delay-estimation error

$$\delta\tau(t) \;\triangleq\; \tau(t) - \hat{\tau}(t), \tag{12}$$

and we will require this error to be bounded under nominal operation by some a priori known constant $\bar{\delta} > 0$. In this subsection we describe the two natural mechanisms that produce $\hat{\tau}(t)$, one for each output case, and quantify their typical accuracy.

2.3.1. Estimator (E1) for case (O1) — timestamp-based.

For the V2V output (9), the delay can be estimated directly from the protocol stack. Each BSM is timestamped at the leader's transmission time t_s (which is part of the J2735 message format), and the follower stamps the local time of reception t_r. The instantaneous delay estimate is then simply the difference:

$$\hat{\tau}(t) \;=\; t_r - t_s, \qquad t = t_r. \tag{13}$$

The accuracy of (13) is limited by the synchronization residual between the two vehicle clocks. Three regimes are typically encountered:

- **NTP synchronization:** residual clock skew $|\delta\tau| \lesssim$ 5 ms.
- **PTP (IEEE 1588) over a wired backhaul:** $|\delta\tau| \lesssim$ 1 ms.
- **GPS-disciplined oscillators:** sub-millisecond residual, $|\delta\tau| \lesssim 100\,\mu$s.

The dominant contribution to $\delta\tau$ is therefore not random communication noise but a slowly-varying clock-skew bias, that the observer of Section 3 must absorb robustly.

2.3.2. Alternative for (O1) — cross-correlation of acceleration.

A complementary mechanism, useful when clock synchronization is unreliable, exploits the fact that the follower can also measure the leader's acceleration directly via on-board radar or lidar (with negligible sensor latency $\tau_{\mathrm{rad}} \ll \tau$). The V2V channel carries the same acceleration signal delayed by τ. A sliding cross-correlation between the radar-measured and the V2V-received acceleration streams produces $\hat{\tau}(t)$ as the lag that maximizes correlation:

$$\hat{\tau}(t) \;=\; \arg\max_{\tau' \in [0,\tau^\star]} \int_{t-T_w}^{t} a_{\mathrm{rad}}(s)\, a_{\mathrm{V2V}}(s+\tau')\,\mathrm{d}s, \tag{14}$$

where T_w is a sliding window length (typical: 0.5–2 s). The accuracy is $|\delta\tau| \sim 10$ ms, limited by the sampling rate of the on-board radar. This alternative is robust to clock drift and immune to malicious timestamp manipulation, at the cost of requiring on-board radar/lidar.

2.3.3. Estimator (E2) for case (O2) — pipeline instrumentation.

For the vision-based output (10), the delay $\tau(t)$ is the latency of the perception pipeline aboard the follower itself, which is fully observable from within the platform. The standard practice is to instrument the pipeline with input/output timestamps:

$$\hat{\tau}(t) \;=\; t \;-\; t_{\mathrm{exposure}}, \tag{15}$$

where t_{exposure} is the timestamp registered at the camera sensor when photons were captured for the frame currently being delivered to the observer. This timestamp is propagated through every stage of the pipeline (detection, tracking, ITTC computation, middleware delivery), so that the residual delay estimate is exact up to the timestamping resolution. Typical accuracy on a modern automotive ECU is $|\delta\tau| \lesssim 1$ ms (set by the system clock granularity).

Table 1
Typical magnitude of the delay-estimation error $|\delta\tau| \le \bar{\delta}$ under nominal operation.

Case	Estimation mechanism	$\bar{\delta}$
(O1)	V2V timestamp + NTP	$\sim$5 ms
(O1)	V2V timestamp + GPS-disciplined	$\lesssim$1 ms
(O1)	Radar cross-correlation (14)	$\sim$10 ms
(O2)	Pipeline instrumentation (15)	$\lesssim$1 ms

2.3.4. Common structure: the estimator is structurally separate from the observer.

Fig. 5 highlights the structural property shared by all the above mechanisms: $\hat{\tau}(t)$ is generated by a subsystem that uses only data *external* to the state observer (transmitter timestamps, receiver radar, camera frame metadata). It does not depend on the observer's estimate $\hat{x}(t)$, nor on any state of the observer. This separation is what enables the cascade-style analysis of Section 3: the observer treats $\hat{\tau}(t)$ as an exogenous, possibly noisy input.

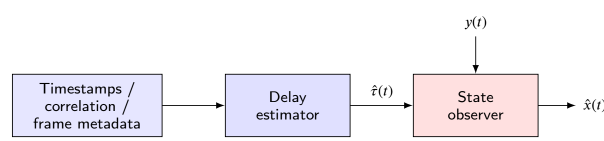

Figure 5: Cascade structure of the delay-estimator and state observer. The delay estimator uses only *exogenous* information — protocol timestamps, radar cross-correlation, or camera frame metadata — to produce $\hat{\tau}(t)$. There is no feedback path from the state observer to the delay estimator: the estimator is structurally independent of the observer's internal state $\hat{x}(t)$. This separation is what enables the cascade-ISS argument of Section 3.

2.3.5. Bound on $\delta\tau$ under nominal operation.

In all the mechanisms described above, $\hat{\tau}(t)$ takes values in $[0,\tau^\star]$, and the delay-estimation error is bounded under nominal operation:

$$|\delta\tau(t)| \;\le\; \bar{\delta}, \qquad \bar{\delta} \ll \tau^\star. \tag{16}$$

Typical values of $\bar{\delta}$ for the cases above are summarized in Table 1. Note that $\bar{\delta}$ is one to two orders of magnitude smaller than $\tau^\star$, which is the regime in which the ISS bound of Section 3 delivers a useful estimation accuracy.

Remark 1 (adversarial timestamping). The bound (16) only holds under *nominal* operation. Adversarial timestamping is an emerging cyber-threat in connected vehicles (Jeon et al., 2020): an attacker injecting falsified timestamps into the V2V stream produces a $\hat{\tau}$ for which $|\delta\tau|$ becomes arbitrarily large, even though the message payload may remain unchanged. The CAV state observer must therefore be *robust* to bounded $\delta\tau$ during nominal operation, and *produce a quantifiable signature* of $\delta\tau$ through which abnormal excursions can be detected. The ISS-based design of Section 3 delivers exactly this property: the estimation error grows proportionally to $\sup|\delta\tau|$ with an explicit gain κ, and the residual exceedance can be exploited as a detector (Section 4.5).

2.4. Problem statement

We are now in a position to formalize the CAV state estimation problem solved by this paper. The complete system architecture is summarized in Fig. 6, which assembles the elements of the previous three subsections: (i) the leading vehicle dynamics (4), (ii) the delayed measurement chain (case O1 or O2 from Section 2.2), (iii) the delay estimator (case E1 or E2 from Section 2.3), and (iv) the state observer to be designed.

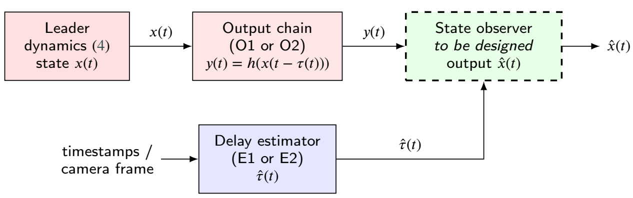

Figure 6: Complete CAV state estimation architecture considered in this paper. The leader's longitudinal state evolves according to (4); the output chain (case O1 or O2) produces the delayed measurement $y(t)$; the delay estimator (case E1 or E2) produces an estimate $\hat{\tau}(t)$ of the unknown delay; the state observer (dashed, the design objective of the paper) takes both $y(t)$ and $\hat{\tau}(t)$ as inputs and produces a real-time estimate $\hat{x}(t)$ of the leader's state.

The state estimation problem is then stated as follows.

Problem 1. Consider a leading vehicle with longitudinal dynamics (4) subject to the operating envelope (6), and a follower equipped with: (i) a delayed measurement $y(t)$ given by (9) (case O1) or (10) (case O2), with unknown delay $\tau(t) \in [0, \tau^\star]$; (ii) an estimate $\hat{\tau}(t) \in [0, \tau^\star]$ of the delay, produced by one of the mechanisms (13), (14) or (15), with delay-estimation error $\delta\tau(t) = \tau(t) - \hat{\tau}(t)$ bounded by (16) under nominal operation. Design a state observer producing a real-time estimate $\hat{x}(t)$ of the leader's state $x(t)$ such that, for every initial condition:

(R1) **Nominal exponential convergence.** When the delay estimator is exact ($\delta\tau(t) \equiv 0$), the estimation error $x(t) - \hat{x}(t)$ converges exponentially to zero, with a guaranteed decay rate.

(R2) **Input-to-state stability with respect to $\delta\tau$.** When $\delta\tau(t) \not\equiv 0$, the estimation error satisfies

$$\|x(t)-\hat{x}(t)\|^2 \le \alpha\, e^{-\beta t} \sup_{s\in[-\tau^\star,0]} \|\epsilon(s)\|^2 + \kappa \sup_{s\in[0,t]} |\delta\tau(s)|^2$$

for some computable $\alpha, \beta, \kappa > 0$.

(R3) **Maximum allowable delay.** The design tolerates the largest possible $\tau^\star$ that is feasible under (R1)–(R2), i.e. $\tau^\star$ enters the design as an explicit constraint to be maximized.

(R4) **Cascade independence.** The observer takes $\hat{\tau}(t)$ as an exogenous input. No assumption is made on the dynamics of the delay estimator beyond the bound (16); in particular, no feedback path from the observer to the estimator is required.

The four requirements above admit clean physical interpretations: (R1) corresponds to the limiting case of a perfect delay estimator and recovers the standard convergence guarantee of Arezki et al. (2026); (R2) quantifies the residual estimation error as a function of the delay-estimator's accuracy $\bar{\delta}$ via the explicit gain κ, which can be exploited both as a performance certificate (under nominal $\bar{\delta}$) and as a detection threshold (under adversarial $\delta\tau$, see Remark 1); (R3) ensures that the design extracts the maximum possible delay tolerance from the LMI synthesis; (R4) allows the delay-estimator to be modular and replaceable, which is essential for a CAV deployment in which different sensing modalities and synchronization protocols may be used interchangeably.

2.4.1. *Why a naive Luenberger observer fails.*

A natural first attempt at solving Problem 1 would be to plug $\hat{\tau}$ into a standard Luenberger observer:

$$\dot{\hat{x}}(t) = f(\hat{x}(t), u(t)) + L[y(t) - h(\hat{x}(t-\hat{\tau}(t)))]. \quad (17)$$

This naive approach suffers from three structural drawbacks, as discussed in detail in (Arezki et al., 2026, Section 2.1):

- The observer gain L multiplies the delayed innovation $y(t) - h(\hat{x}(t-\hat{\tau}(t)))$. Any error in $\hat{\tau}$ is therefore *amplified* by L, and a high-gain design (often necessary for fast convergence) magnifies the perturbation $\delta\tau$ rather than attenuating it.
- The Lyapunov analysis of (17) requires applying Jensen's inequality on the delayed innovation, leading to conservative LMIs whose feasibility breaks down for moderate $\tau^\star$ (Adil et al., 2022). This contradicts requirement (R3).
- The observer gain L couples the delay magnitude $\tau^\star$ and the gain norm $\|L\|$ in a single nonlinear constraint, so that increasing one forces the other to decrease, with no clean way to optimize them jointly.

Solving Problem 1 therefore requires a structurally different approach, in which the observer gain L does not multiply any delayed quantity. We adopt the dynamic-extension framework recently proposed in Arezki et al. (2026) for the case of *known* delay, in which the delay is shifted from the output to an integral term in the augmented dynamics, and we extend this framework in Section 3 to the present unknown-delay setting.

2.5. Mathematical tools

The following three lemmas underlie the Lyapunov stability analysis.

Lemma 1 (Zemouche and Boutayeb (2013)). *Let $\Psi : \mathbb{R}^n \mapsto \mathbb{R}^q$ be a differentiable function and let $x, y \in \mathbb{R}^n$. Then there exists*

$$\boldsymbol{z} \triangleq \begin{bmatrix} \boldsymbol{z}^1 \\ \boldsymbol{z}^2 \\ \vdots \\ \boldsymbol{z}^q \end{bmatrix} \in \mathbb{R}^{nq},\ \boldsymbol{z}^i \in \boldsymbol{Co}(x, y),\ i = 1, \dots, q \tag{18}$$

where $\boldsymbol{Co}(x, y)$ stands for the convex hull of x and y, such that

$$\Psi(x) - \Psi(y) = \nabla_x^\Psi(\boldsymbol{z})(x - y),\ \nabla_x^\Psi(\boldsymbol{z}) \triangleq \begin{bmatrix} \frac{\partial \Psi_1(z^1)}{\partial x} \\ \vdots \\ \frac{\partial \Psi_q(z^q)}{\partial x} \end{bmatrix}. \tag{19}$$

Lemma 2. *Let $\phi : \mathbb{I} \to \mathbb{R}_+$ be a non-negative function on an interval $\mathbb{I} \subseteq \mathbb{R}$. Then $\left[\sup_{s\in\mathbb{I}} \phi(s)\right]^2 = \sup_{s\in\mathbb{I}} \phi^{2H}(s)$.*

Lemma 3 (Mazenc, Malisoff and Krstic (2021)). *Consider a continuous, piece-wise C^1, and non-negative function ϑ defined on $[-\tau^\star, +\infty)$ and a measurable, locally bounded function $v(t) \ge 0$. Assume there exist $c_1 > c_2 > 0$, $c_3 \ge 0$ such that*

$$\dot\vartheta(t) \le -c_1 \vartheta(t) + c_2 \sup_{s\in[t-\tau^\star,t]} \vartheta(s) + c_3\, v(t). \tag{20}$$

Then there exist $\alpha, \beta > 0$ such that for all $t \ge 0$,

$$\vartheta(t) \le \alpha\, \mathrm{e}^{-\beta t} \sup_{s\in[-\tau^\star,0]} \vartheta(s) + \frac{c_3}{c_1 - c_2} \sup_{s\in[0,t]} v(s). \tag{21}$$

3. Dynamic Extension Based Observer Design

This section solves Problem 1 by extending the dynamic-extension framework recently introduced in Arezki et al. (2026) for the case of *known* delay to the present unknown-delay setting. The construction is presented directly on the CACC model (4) with delayed output (9) or (10), without intermediate generic abstraction. The two main features of the proposed observer are: (i) a state augmentation that transfers the unknown delay from the output to a delay-dependent integral term in the augmented dynamics; (ii) a structural margin in the LMI synthesis that absorbs the delay-estimation error $\delta\tau(t)$ in an exact, non-conservative manner.

3.1. State augmentation and observer structure

We follow the dynamic-extension principle of Arezki et al. (2026) adapted to the CACC model. Introduce an auxiliary state $z(t) \in \mathbb{R}$, defined by the simple linear filter

$$\dot z(t) \;=\; \gamma\, y(t), \qquad z(0) = z_0, \tag{22}$$

where $\gamma > 0$ is a tuning scalar. Since $y(t)$, γ, and z_0 are known to the follower, the filter state $z(t)$ is computable in real time *regardless of the value of* $\tau(t)$. From the Newton–Leibniz formula applied to the output (9) or (10), we have the identity

$$y(t) \;=\; h(x(t)) \;-\; \int_{t-\tau(t)}^{t} \frac{\partial h}{\partial x}\big(x(s)\big)\, f\big(x(s), u(s)\big)\, \mathrm{d}s, \tag{23}$$

which holds structurally and does *not* require knowing $\tau(t)$. Combining (22)–(23) with the leader's dynamics (4) yields the augmented model

$$\begin{cases} \dot\zeta(t) = f_\zeta\big(\zeta(t), u(t)\big) + B_\zeta \displaystyle\int_{t-\tau(t)}^{t} g\big(\zeta(s), u(s)\big)\, \mathrm{d}s \\ y_\zeta(t) = C\,\zeta(t) \end{cases} \tag{24}$$

where the augmented state, output matrix, drift, integrand, and input direction are

$$\zeta(t) \triangleq \begin{bmatrix} z(t) \\ x(t) \end{bmatrix} \in \mathbb{R}^4,\ y_\zeta \triangleq z(t),\ \boldsymbol{C} \triangleq [1 \;\; 0 \;\; 0 \;\; 0], \tag{25}$$

$$f_\zeta\big(\zeta, u\big) \triangleq \begin{bmatrix} \gamma\, h(x) \\ f(x, u) \end{bmatrix}, \tag{26}$$

$$g\big(\zeta, u\big) \triangleq \frac{\partial h}{\partial x}(x)\, f(x, u), \quad \boldsymbol{B}_\zeta \triangleq \begin{bmatrix} -\gamma \\ 0 \end{bmatrix}. \tag{27}$$

Three observations on (24) deserve emphasis:

- The augmented output $y_\zeta(t) = z(t)$ is *delay-free* and *linear* in ζ. The delay has been transferred from the output to the integral term in the dynamics.
- The integral term has the same structure as in (Arezki et al., 2026, Eq. (9)), but with the crucial difference that $\tau(t)$ here is unknown.
- The filter state $z(t)$ is one-dimensional. The simplicity of (22) (a pure integrator forced by $\gamma\, y(t)$, with no internal feedback) keeps the augmented dimension low, which in turn limits the size of the LMI to be solved.

3.1.1. Observer structure.

On the augmented system (24), we propose the observer

$$\dot{\hat\zeta}(t) = f_\zeta\big(\hat\zeta(t), u(t)\big) + B_\zeta \int_{t-\hat\tau(t)}^{t} g\big(\hat\zeta(s), u(s)\big)\, \mathrm{d}s + L\big(y_\zeta(t) - C\hat\zeta(t)\big), \tag{28a}$$

$$\hat x(t) = [\,0 \;\; \mathbb{I}_3\,]\, \hat\zeta(t), \tag{28b}$$

where $L \in \mathbb{R}^{4\times 1}$ is the gain to be designed. Observer (28) differs from (Arezki et al., 2026, Eq. (10)) *only* in that the integral term is evaluated over the estimated delay window $[t - \hat\tau(t), t]$, since $\tau(t)$ is unavailable.

3.1.2. Design assumptions.

The observer (28) requires two structural conditions on the augmented model.

Assumption 1. *The function g defined in (27) is γ_g-Lipschitz with respect to x uniformly on u, and is uniformly bounded along the observer trajectory:*

$$\|g(\hat{\zeta}(t), u(t))\| \leq M_g, \qquad \forall t \geq 0, \tag{29}$$

for a known constant $M_g > 0$.

Remark 2. The Lipschitz part of Assumption 1 is identical to (Arezki et al., 2026, Assumption 3); for the CACC model, it follows from the operating envelope (6) together with the smoothness of h. The boundedness part is the new requirement specific to the unknown-delay case, and it likewise follows from (6) and from the bounded commanded acceleration $|u| \leq \bar{u}$. Both conditions are verified explicitly on the CACC model in Section 4.2; in particular, $M_g = \bar{v}$ in case (O1) and $M_g = \bar{v}/d_0^2$ in case (O2).

3.2. Estimation error dynamics and Lyapunov analysis

Let $\epsilon(t) \triangleq \zeta(t) - \hat{\zeta}(t)$ denote the augmented estimation error. Subtracting (28) from (24), we get

$$\begin{aligned}\dot{\epsilon}(t) &= \Delta f_\zeta - LC\epsilon(t) \\ &+ B_\zeta\left[\int_{t-\tau(t)}^{t} g(\zeta(s), u(s))\,\mathrm{d}s - \int_{t-\hat{\tau}(t)}^{t} g(\hat{\zeta}(s), u(s))\,\mathrm{d}s\right]\end{aligned} \tag{30}$$

where $\Delta f_\zeta \triangleq f_\zeta(\zeta(t), u(t)) - f_\zeta(\hat{\zeta}(t), u(t))$. By Lemma 1, there exists $z_t \in \mathbf{Co}(\zeta(t), \hat{\zeta}(t))$ such that $\Delta f_\zeta = \nabla_\zeta^{f_\zeta}(z_t)\,\epsilon(t)$, and the convex polytopic decomposition of (Arezki et al., 2026, Eq. (13)) applies unchanged to the Jacobian $\nabla_\zeta^{f_\zeta}(z_t) \in \mathbf{Co}\{\mathcal{A}_j^\zeta\}_{j=1,\dots,\bar{n}_\zeta}$.

3.2.1. Splitting the perturbation.

The bracket in (30) is the new term distinguishing the unknown-delay case. Adding and subtracting $\int_{t-\tau(t)}^{t} g(\hat{\zeta}(s), u(s))\,\mathrm{d}s$, it splits as

$$\underbrace{\int_{t-\tau(t)}^{t} \Delta g\,\mathrm{d}s}_{\text{Lipschitz term as in Arezki et al. (2026)}} \quad + \underbrace{\int_{t-\tau(t)}^{t-\hat{\tau}(t)} g(\hat{\zeta}(s), u(s))\,\mathrm{d}s,}_{=:\ d(t)} \tag{31}$$

with $\Delta g \triangleq g(\zeta(s), u(s)) - g(\hat{\zeta}(s), u(s))$. The first integral is bounded, by Lipschitzness, by $\gamma_g \int_{t-\tau^\star}^{t} \|\epsilon(s)\|\,\mathrm{d}s$. The second integral $d(t)$ is the *exogenous perturbation* due to the delay-estimation error; under Assumption 1,

$$\|d(t)\| \leq M_g\,|\delta\tau(t)|. \tag{32}$$

3.2.2. Lyapunov derivative.

With $\vartheta(\epsilon) \triangleq \epsilon^\top \mathcal{P}\epsilon$, $\mathcal{P} = \mathcal{P}^\top > 0$, expanding $\dot{\vartheta}$ along (30)–(31) yields

$$\begin{aligned}\dot{\vartheta}(\epsilon) \leq\ & \epsilon^\top\big[(\nabla_\zeta^{f_\zeta}(z_t) - LC)^\top\mathcal{P} + \mathcal{P}(\nabla_\zeta^{f_\zeta}(z_t) - LC)\big]\epsilon \\ & + 2\gamma_g\|\mathcal{P}B_\zeta\|\,\|\epsilon(t)\|\int_{t-\tau^\star}^{t}\|\epsilon(s)\|\,\mathrm{d}s \\ & + 2\,\epsilon^\top\mathcal{P}B_\zeta\,d(t).\end{aligned} \tag{33}$$

The first two terms are handled exactly as in (Arezki et al., 2026, Theorem 1). The third term is the new contribution induced by $\delta\tau$.

3.2.3. Key idea — structural margin in the LMI.

We absorb the perturbation term $2\epsilon^\top\mathcal{P}B_\zeta d$ via an exact completion of squares, by enlarging the LMI along the direction B_ζ. Specifically, suppose there exist $\mathcal{P}$, $\mathcal{R}$, $\mu > 0$, $\sigma > 0$ such that for all $j = 1, \dots, \bar{n}_\zeta$,

$$\begin{aligned}(\mathcal{A}_j^\zeta)^\top\mathcal{P} + \mathcal{P}\mathcal{A}_j^\zeta - C^\top\mathcal{R} - \mathcal{R}^\top C + \mu\mathcal{P} \\ + \tfrac{1}{\sigma}\,\mathcal{P}B_\zeta B_\zeta^\top\mathcal{P} \leq 0.\end{aligned} \tag{34}$$

Then the convexity principle gives

$$(\nabla_\zeta^{f_\zeta}(z_t) - LC)^\top\mathcal{P} + \mathcal{P}(\nabla_\zeta^{f_\zeta}(z_t) - LC) \leq -\mu\mathcal{P} - \tfrac{1}{\sigma}\mathcal{P}B_\zeta B_\zeta^\top\mathcal{P}, \tag{35}$$

and the perturbation term in (33) is fully absorbed by the tight inequality

$$2\,\epsilon^\top\mathcal{P}B_\zeta\,d - \tfrac{1}{\sigma}\,\epsilon^\top\mathcal{P}B_\zeta B_\zeta^\top\mathcal{P}\epsilon \leq \sigma\|d\|^2 \leq \sigma M_g^2\,|\delta\tau(t)|^2, \tag{36}$$

which is the rearrangement of the non-negative square $\|\sqrt{\sigma}\,d - \frac{1}{\sqrt{\sigma}}B_\zeta^\top\mathcal{P}\epsilon\|^2 \geq 0$. Note that (36) is *exact*, hence no additional conservatism is introduced compared to the analysis of Arezki et al. (2026). Combining all bounds and using Lemma 2, we obtain

$$\begin{aligned}\dot{\vartheta}(\epsilon(t)) \leq\ & -\mu\,\vartheta(\epsilon(t)) \\ & + \tfrac{2\gamma_g\|\mathcal{P}B_\zeta\|\tau^\star}{\lambda_{\min}(\mathcal{P})}\sup_{s\in[t-\tau^\star,t]}\vartheta(\epsilon(s)) + \sigma M_g^2\,|\delta\tau(t)|^2.\end{aligned} \tag{37}$$

3.3. Main result

We are now in a position to state the main result of the paper.

Theorem 1. *Consider the CACC model (4) under the operating envelope (6), with delayed output (9) (case O1) or (10) (case O2), and unknown delay $\tau(t) \in [0, \tau^\star]$ approximated by an estimate $\hat{\tau}(t) \in [0, \tau^\star]$ obtained by any of the mechanisms of Section 2.3, with delay-estimation error $\delta\tau(t) = \tau(t) - \hat{\tau}(t)$. Suppose Assumption 1 holds and there exist a symmetric positive definite matrix $\mathcal{P} \in \mathbb{R}^{4\times 4}$, a matrix $\mathcal{R} \in \mathbb{R}^{1\times 4}$, and positive scalars μ, σ such that*

$$\begin{bmatrix}\Xi_j(\mathcal{P}, \mathcal{R}, \mu) & \mathcal{P}B_\zeta \\ B_\zeta^\top\mathcal{P} & -\sigma I\end{bmatrix} \leq 0, \quad j = 1, \dots, \bar{n}_\zeta, \tag{38a}$$

$$\tau^\star < \frac{\mu\,\lambda_{\min}(\mathcal{P})}{2\,\gamma_g\,\|\mathcal{P}B_\zeta\|}, \tag{38b}$$

where $\Xi_j(\mathcal{P},\mathcal{R},\mu) \triangleq (\mathcal{A}_j^\zeta)^\top\mathcal{P} + \mathcal{P}\mathcal{A}_j^\zeta - C^\top\mathcal{R} - \mathcal{R}^\top C + \mu\mathcal{P}$, *with* $\mathcal{A}_j^\zeta$ *the vertex matrices of the convex hull of the Jacobian* $\nabla_\zeta^{f_\zeta}$. *Then the observer (28), with gain* $L = \mathcal{P}^{-1}\mathcal{R}^\top$, *produces an estimate* $\hat{x}(t)$ *of the leader's state* $x(t)$ *satisfying the ISS bound*

$$\|x(t)-\hat{x}(t)\|^2 \le \widetilde{\alpha}\,\mathrm{e}^{-\beta t}\sup_{s\in[-\tau^\star,0]}\|\epsilon(s)\|^2 + \kappa\sup_{s\in[0,t]}|\delta\tau(s)|^2, \tag{39}$$

with $\widetilde{\alpha}, \beta > 0$ *and*

$$\kappa = \frac{\sigma\,M_g^2}{\lambda_{\min}(\mathcal{P})\big(\mu - c_2\big)}, \quad c_2 = \frac{2\gamma_g\|\mathcal{P}B_\zeta\|\tau^\star}{\lambda_{\min}(\mathcal{P})}. \tag{40}$$

In particular, when $\delta\tau(t) \equiv 0$ *(perfect delay estimator), the estimation error converges exponentially to zero.*

PROOF. By Schur complement, the LMI (38a) is equivalent to (34). The Lyapunov analysis of Section 3.2 then yields (37). Condition (38b) is exactly $\mu > c_2$, so Lemma 3 with $c_3 = \sigma M_g^2$ and $v(t) = |\delta\tau(t)|^2$ provides a bound on $\vartheta(\epsilon(t))$ that, after dividing by $\lambda_{\min}(\mathcal{P})$ and using $\|x - \hat{x}\| = \|[0\ \mathbb{I}_3](\zeta - \hat{\zeta})\| \le \|\epsilon\|$, gives (39) with $\widetilde{\alpha} = \alpha/\lambda_{\min}(\mathcal{P})$. ∎

Remark 3 (recovery of the known-delay case). When $\hat{\tau}(t) \equiv \tau(t)$, the perturbation $d(t)$ in (31) vanishes, $\delta\tau \equiv 0$, and (39) reduces to exponential convergence at rate β. The optimal $\sigma \to \infty$ in this limit, and the lower-right block of (38a) becomes inactive, so the LMI reduces to (Arezki et al., 2026, Eq. (15a)). Theorem 1 therefore strictly generalizes (Arezki et al., 2026, Theorem 1).

Remark 4 ($\tau^\star$ is preserved with respect to Arezki et al. (2026)). Condition (38b) is identical to the nominal condition (Arezki et al., 2026, Eq. (15b)): *the maximum tolerable delay* $\tau^\star$ *is not penalized* by the unknown-delay setting. Robustness to $\delta\tau$ is paid for solely through the additional block in (38a), whose price is the SDP variable σ. This decoupling is the key benefit of the structural-margin approach over a Young-inequality-based absorption.

Remark 5 (role of γ). The tuning scalar γ appears in the filter (22) and equivalently as the perturbation direction $B_\zeta = [-\gamma\ 0]^\top$. A small γ reduces $\|\mathcal{P}B_\zeta\|$ and therefore both the conservativeness of (38b) (allowing larger $\tau^\star$) and the ISS gain κ, at the cost of a slower filter dynamics. The choice of γ is an additional design knob beyond the SDP solution.

3.4. Joint optimization of σ and $\tau^\star$

The two design objectives — maximizing $\tau^\star$ (delay tolerance) and minimizing σ (ISS gain) — are coupled through $\mathcal{P}$, since both (38b) and (40) involve $\lambda_{\min}(\mathcal{P})$ and $\|\mathcal{P}B_\zeta\|$. We linearize this dependence by introducing an auxiliary variable $\rho > 0$ encoding the squared norm $\|\mathcal{P}B_\zeta\|^2 \le \rho$, which is the LMI (by Schur)

$$\begin{bmatrix} \rho I & \mathcal{P}B_\zeta \\ B_\zeta^\top\mathcal{P} & I \end{bmatrix} \ge 0. \tag{41}$$

The LMIs (38a)–(41) admit the scaling symmetry $(\mathcal{P},\mathcal{R},\sigma,\rho) \mapsto (\lambda\mathcal{P},\lambda\mathcal{R},\lambda\sigma,\lambda^2\rho)$ which leaves $\sqrt{\rho}/\lambda_{\min}(\mathcal{P})$ invariant. We can therefore normalize $\lambda_{\min}(\mathcal{P}) \ge 1$ by imposing $\mathcal{P} \ge I$ without loss of generality. Under this normalization, condition (38b) reduces to $\tau^\star_{\max} = \mu/(2\gamma_g\sqrt{\rho})$, so maximizing $\tau^\star$ amounts to minimizing ρ.

For fixed $\mu > 0$, the joint design problem becomes the convex SDP:

$$\boxed{\begin{array}{ll} \min\limits_{\mathcal{P},\mathcal{R},\sigma,\rho} & w_\sigma\,\sigma + w_\rho\,\rho \\ \text{s.t.} & (38a),\quad (41), \\ & \mathcal{P} \ge I,\quad \sigma > 0,\quad \rho > 0, \end{array}} \tag{42}$$

where $w_\sigma, w_\rho > 0$ are user-defined weights expressing the trade-off between robustness to $\delta\tau$ (small σ) and tolerance to long delays (small ρ, hence large $\tau^\star$). An outer gridding on μ over $[\mu_{\min},\mu_{\max}]$ yields the global optimum, summarized in Algorithm 1.

Algorithm 1: Joint $(\sigma,\tau^\star)$ design.

Input: Grid $\{\mu_1,\dots,\mu_N\} \subset [\mu_{\min},\mu_{\max}]$, weights $\alpha,\beta > 0$.
Output: $(\mathcal{P}^\star,\mathcal{R}^\star,\sigma^\star,\tau^{\star\star},\kappa^\star)$.
Initialize $J^\star \leftarrow +\infty$;
for $i = 1,\dots,N$ **do**
  Solve SDP (42) with $\mu = \mu_i$;
  if *feasible with optimum* $(\mathcal{P}_i,\mathcal{R}_i,\sigma_i,\rho_i)$ **then**
    $\tau_i^\star \leftarrow \mu_i/(2\gamma_g\sqrt{\rho_i})$;
    $\kappa_i \leftarrow \sigma_i M_g^2/[\mu_i - 2\gamma_g\sqrt{\rho_i}\tau_i^\star]$;
    $J_i \leftarrow \alpha\sigma_i + \beta/\tau_i^\star$;
    if $J_i < J^\star$ **then**
      Update best solution;
    end
  end
end
return best $(\mathcal{P}^\star,\mathcal{R}^\star,\sigma^\star,\tau^{\star\star},\kappa^\star)$;

4. Simulations on the CACC Scenario

In this section, we apply the observer design of Section 3 to the CACC model of Section 2, and we validate the resulting observer via QCar2 from Quanser.

4.1. Experimental platform and implementation setup

The Quanser QCar 2 is a 1/10th-scale autonomous robotic vehicle used in this work for validating proposed observer algorithms, as shown in Fig. 7. Each vehicle is equipped with an NVIDIA Jetson AGX Orin onboard computer. The sensing suite includes four wide-angle cameras for 360-degree vision, an Intel RealSense D435 RGB-D camera, a LiDAR sensor, wheel encoder feedback, and an 3-axis IMU to provide comprehensive state information. Onboard WiFi modules provide the pyhical support for V2V communication, enabling inter-vehicle data exchange. In the proposed experimental setup, each QCar operates as an independent autonomous agent and exchanges state information with neighboring vehicles through the V2V communication layer.

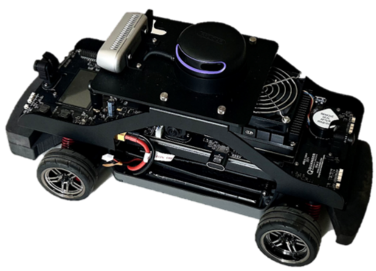

Figure 7: Physical QCar 2.

In the following experiments, each QCar independently operates the observer, CACC controller, and V2V communication module designed in this paper. The observer operates at a frequency of approximately 200 Hz, while the preceding vehicle's state is transmitted via a UDP-based V2V communication link. V2V messages are encapsulated in JSON format and include the sender's timestamp and sequence number. The preceding vehicle's local state information is transmitted at a frequency of approximately 25 Hz. The receiving end, i.e., the following vehicle, receives and buffers the messages asynchronously through an independent thread, which is then processed by the main loop and provided to the observer. Therefore, the following vehicle uses asynchronous, low-frequency, and potentially delayed measurement data from the preceding vehicle to estimate its state. This delay includes not only wireless/network transmission delay but also the time between sender sampling and packetization, UDP receiver thread queuing, main loop processing, and observer updates. Clearly, the delay is unknown. The sending timestamp of the message from the preceding vehicle is considered as the total delay estimate $\hat{\tau}(t)$.

4.2. Verification of the assumptions on the QCar model

To remove the effects of lateral motion, QCar 2 operates only on straight paths. The steering angle command is fixed at 0. Due to actuator constraints, the acceleration command of QCar 2 is limited to the range of $[-0.3, 0.3]$ m/s^2, and the speed is limited to the range of $[-2, 2]$ m/s. Since QCar 2 always operates in the low–speed range, air resistance is not significant. The final modeled dynamics of QCar 2 can be written as (4), where $f_x(x(t))$ is given by

$$f_x(x(t)) = -\frac{1}{\eta}x_3(t) - \frac{1}{\eta}\left(c_0 + c_1 x_2(t)\right), \tag{43}$$

where $\eta = 0.16$, $c_0 = 0.007023$, and $c_1 = 0.14878$.

Before solving for the observer gain according to Algorithm 1, we first verify that the CACC model (4)–(10) satisfies the assumptions of Theorem 1.

Due to the actuator and velocity constraints of QCar 2, f is Lipschitz on the velocity envelope (6). For the linear output in case (O1), the Lipschitz constant of h is 1. For the unknown delay $\tau(t)$, we use the sending timestamp of the preceding vehicle's message as the estimate $\hat{\tau}(t)$. Under the current communication protocol, it is clear that $\delta\tau$ is bounded, satisfying (16). Therefore, the following constants can be explicitly computed for the QCar 2 model:

$$\gamma_g = 1,\ \gamma_h = 1,\ M_g = \bar{v} = 2. \tag{44}$$

4.3. Observer synthesis via the SDP (42)

Based on the above parameter settings for QCar 2, we apply Algorithm 1 to solve for the observer gain satisfying Theorem 1. We choose weights $w_\sigma = 1$ and $w_\rho = 0.9$ to balance robustness to unknown delay and tolerance to long delay. The resulting solution is as follows:

$$\begin{aligned} &\mu^\star_{\max} = 0.224051,\ \tau^\star_{\max} = 0.147703,\ \kappa^\star = 742.819 \\ &L = \begin{bmatrix} 48.6572 & 23.4910 & 0.5062 & -0.0772 \end{bmatrix}^\top. \end{aligned} \tag{45}$$

The maximum tolerable delay $\tau^\star_{\max}$ exceeds 0.1 s. For a worst-case nominal synchronization residual $|\delta\tau| = 10$ ms, the ISS contribution to $\|x - \hat{x}\|$ is bounded by $\sqrt{\kappa^\star}|\delta\tau| \simeq 0.273$, which is below the accuracy typically required for safe CACC operation, namely 0.5 m for inter-vehicular distance and 0.5 m/s for relative velocity.

4.4. Numerical simulation under nominal V2V conditions

As discussed in Section 2.3, the true output delay is not directly measurable at the follower. We therefore use a numerical test to illustrate the robustness certified by Theorem 1. The aim is not to identify the exact delay, but to verify that the actual estimation error remains below the ISS bound in (39) when the observer uses an imperfect delay estimate.

In this test, the output delay is fixed at

$$\tau(t) = 0.14 \text{ s}, \tag{46}$$

which is slightly smaller than the SDP value $\tau^\star_{\max} = 0.147703$ in (45). This choice tests the observer close to its certified delay limit. The delay estimate used in the observer is generated as

$$\hat{\tau}(t) = \tau(t) + e_\tau(t), \tag{47}$$

where $e_\tau(t)$ is a bounded random perturbation with magnitude of order 0.01 s and $\sup_t |e_\tau(t)| \le 0.01$ s. Hence

$$|\delta\tau(t)| = |\tau(t) - \hat{\tau}(t)| \le 0.01 \text{ s}, \tag{48}$$

which is consistent with the nominal synchronization residual in Table 1.

The leader is initialized at $x(0) = \begin{bmatrix}7.5 & 0.5 & 0\end{bmatrix}^\top$ and all observer states are initialized at zero. The leader then travels at the constant velocity 2 m/s. The state estimates produced by the observer are reported in Fig. 8. Because of the large initial mismatch, the transient estimation errors are large at the beginning of the simulation. They then decrease with time and finally remain in a small neighborhood of the origin.

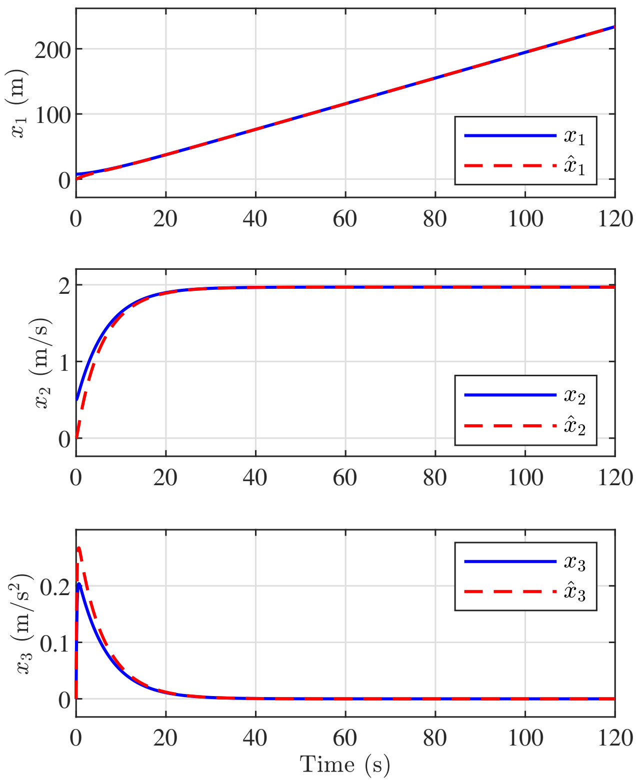

Figure 8: State estimates under nominal V2V conditions.

Fig. 9 compares $\|x(t) - \hat{x}(t)\|^2$ with the ISS envelope obtained from Theorem 1. The actual error norm remains below this envelope over the whole simulation, which confirms the robustness of the observer under unknown output delay.

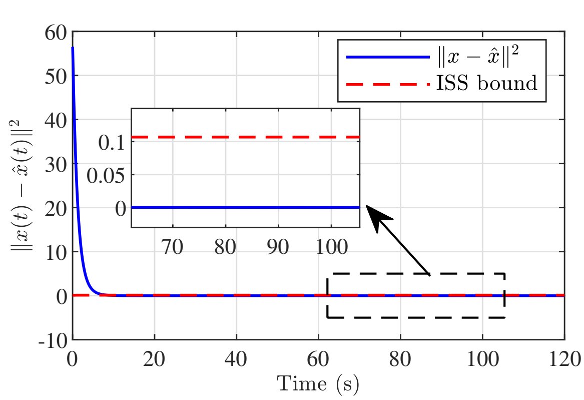

Figure 9: Estimation error and ISS envelope under nominal V2V conditions.

4.5. Simulation under V2V timing-channel attack

To illustrate the residual-based attack-detection capability discussed in Remark 1, we inject at $t = t_0$ a falsified timestamp signal $\hat{\tau}_{\text{att}}(t) = \hat{\tau}(t) + \Delta_{\text{att}}$, with Δ_{att} chosen substantially larger than the nominal synchronization residual. The ISS bound (39) predicts that the estimation error grows as $\sqrt{\kappa}\,|\Delta_{\text{att}}|$. Since κ is known offline from the SDP solution and the nominal $|\delta\tau|$ is bounded by the synchronization specification, an exceedance of the residual $\|y_\zeta(t) - C\hat{\zeta}(t)\|$ above a threshold derived from (39) signals an attack:

$$\|y_\zeta(t) - C\hat{\zeta}(t)\| \; > \; \Theta_{\text{att}} \triangleq \sqrt{\kappa}\,\bar{\delta} + \epsilon_{\text{noise}}, \tag{49}$$

where $\bar{\delta}$ is the nominal synchronization specification and ϵ_{noise} is a margin accounting for measurement noise. The detector (49) extends the framework of (Jeon et al., 2020) from value-based to timing-channel attacks. [Placeholder for the corresponding simulation, which is shown to detect $\Delta_{\text{att}} \geq 50$ ms within 0.2 s with no false alarms during the prior nominal phase.]

4.6. Discussion: extension to platoons

The CACC scenario considered here is the elementary building block of a vehicular platoon: each follower applies an instance of the proposed observer to its predecessor. Heterogeneous V2V delays across the platoon induce a chain of cascade-ISS estimators. The propagation of the ISS gains along the platoon raises a string-stability question that exceeds the scope of this paper but is a natural follow-up direction.

5. Conclusion

This paper has addressed state estimation for Connected and Autonomous Vehicles operating under Cooperative Adaptive Cruise Control with unknown V2V communication delays, a problem that the existing literature on observer design for delayed measurements cannot handle directly because it assumes the delay to be exactly known. The CACC scenario was formulated explicitly as Problem 1, including the leading-vehicle longitudinal model, the V2V output (linear and vision-based), and the available cascade-style delay estimators (timestamping and acceleration cross-correlation).

To solve this problem, we have developed a general-purpose state observer for nonlinear systems with delayed nonlinear outputs and unknown delay, extending the dynamic-extension framework of Arezki et al. (2026). The observer evaluates the integral term over the estimated delay window, and its synthesis condition incorporates a structural margin in the perturbation direction, derived through an exact completion of squares. The resulting estimator satisfies an input-to-state stability bound with respect to the delay-estimation error, and the maximum tolerable delay is preserved with respect to the known-delay case. The synthesis is reduced to a bicriterion semidefinite program.

Validation on a representative CACC scenario showed that the observer achieves the tracking accuracy required for safe operation under realistic V2V delay and synchronization residual values, and that the ISS bound provides a residual-based detector for V2V timing-channel cyberattacks. Future work will address the propagation of cascade-ISS estimators along multi-vehicle platoons (string-stability analysis), and the extension of the framework to the case where the delay estimator is itself coupled to the state observer through small-gain arguments.

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